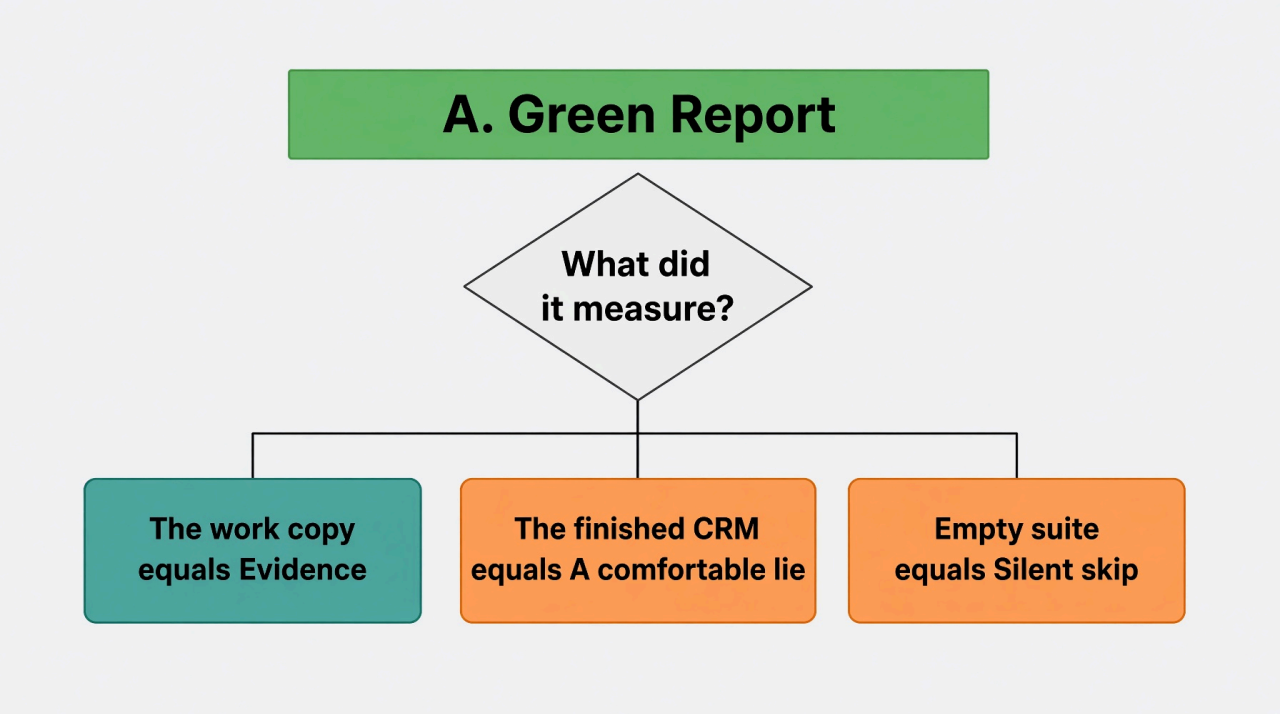
Lab 2. Ticket implementer and the harness

The centre of the workshop. 55 minutes. Never cut.

A ready ticket in. A green rubric out.

25 minutes of typing. Fill `harness.py`. Nothing else.

Work from `labs/lab2_implementer`.



Why this lab exists

The loop you just built will lie to you. Edit forever. Declare victory on red. Stuff the window.

A true story from this repo: seven tests, green on every run, testing the wrong tree. The conftest put the finished answer on `sys.path` ahead of the work copy.

A check that reports success while measuring the wrong thing is worse than no check.

Artifact. A reusable evaluation harness: tests first, prove them red, code until green, judge, gate.

Learning objectives

- Implement `red_gate` so only **new** failing ids count
- Implement `score_attempt` as one call to `rubric.score`
- Implement `run_loop` so Python counts retries, not the model
- Configure two doers with disjoint write scope
- Validate `--doer none` escalates (honesty) and `--doer reference` can pass
- Troubleshoot the stall of computing rubric rows by hand

Starting architecture

Already in this folder: `contract.py`, `rubric.py`, `gates.py`, the stub, prompts.

You create: three function bodies in `harness.py`.

```
orchestrator  (writes nothing, owns budget)
  planner            writes steps.jsonl
  test_implementer   writes tests/**
  code_implementer   writes app/**, denied tests/**
  judge              no write method
          |
          ▼
     red gate  →  ten-row rubric  →  gates.decide
          |
          ▼
     .harness/last-implementer.json
     .harness/receipt.json
```

This is Graph Engineering: each acceptance criterion becomes two nodes in `steps.jsonl`, a test step and a code step. Derived, not generated.

There is no `loops/` package. Do not recreate it.

Prerequisites

```
cd labs/lab2_implementer
```

Pick one tool and paste its prompt:

```
claude -p "$(cat prompts/claude-code.md)"
# or: codex exec "$(cat prompts/codex.md)"
# or: grok -p "$(cat prompts/grok-build.md)"
# or: opencode run "$(cat prompts/opencode.md)"
```

CRM clone from Lab 1 should already sit at `../../work/northwind-field-crm`.

`.claude/settings.json` denies writes to `loops/`, `scripts/`, and `work/**/tests/**`.

The stub. Three functions. The order is the lesson

```
def red_gate(before: RunResult, after: RunResult) -> set[str]:
    raise NotImplementedError("fill me in")

def score_attempt(contract: Contract, **evidence) -> rubric.Score:
    raise NotImplementedError("fill me in")

def run_loop(contract: Contract, budget: int = 3) -> dict:
    raise NotImplementedError("fill me in")
```

Order, from the module docstring:

```
tests first  ->  prove them red  ->  code until green  ->  judge  ->  gate
```

`red_gate`. New failing ids, not any failing ids

A test that already existed and still fails is not proof of a new contract.
A test that passes before any code exists proves nothing.
An empty result must stop the loop.

Body Rick types (and the Deep Agents port ships):

```
def red_gate(before, after) -> set[str]:
    seen = before.junit.passed_ids | before.junit.failed_ids
    return {tid for tid in after.junit.failed_ids if tid not in seen}
```

The DA core names this `implementer._new_test_ids`.

`score_attempt`. One line. Do not compute rows

"The tests passed" is one row of ten. Absent kwargs become failing rows on purpose.

```
def score_attempt(contract: Contract, **evidence) -> rubric.Score:
    return rubric.score(contract=contract, **evidence)
```

That is the whole function. The common stall is writing ten `if` blocks.

#	ROW	EVIDENCE
1	`tests_ran`	junit exists and is non-empty
2	`tests_passed`	failures+errors == 0
3	`red_first`	`red_ids` non-empty
4	`coverage_floor`	line-rate vs floor (80)
5	`criteria_covered`	every criterion has a passing test
6	`steps_done`	no open plan steps
7	`ui_has_e2e`	UI files changed ⇒ e2e green
8	`format_clean`	format task ok
9	`lint_clean`	lint task ok
10	`write_scope`	no violations (and scope was checked)

`run_loop`. Python holds the retries

```
def run_loop(contract, budget: int = 3, ticket_id: str = "T001", doer: str = "reference") -> dict:
    return implementer.run(
        repo=contract.repo, ticket_id=ticket_id, budget=budget, doer=doer,
    )
```

Saturday this folder has no `implementer.py`. Rick types a body that calls `gates.decide` in a `while` loop, or you watch him fill it. The shipped eight-step loop lives in `solutions/sol2_implementer_deep_agents/implementer.py`.

Pass `doer` through. Hardcoding `reference` makes `--doer none` impossible.

Two doers. Scope is a missing path

From `contract.py` defaults:

```
planner            write_allow: steps.jsonl
test_implementer   write_allow: tests/**
code_implementer   write_allow: app/**, src/**    write_deny: tests/**
judge              write_allow: []                write_deny: **
```

Deny always beats allow. Empty allow permits nothing. `Judge` has no `write` method.

The code implementer cannot weaken a test, not because it was told not to, but because it holds no write path to one.

`gates.decide`. Three exits, no fourth

1. Rubric green (`judge_done is None`) → pass
2. Rubric green and final judge agrees → pass
3. Rubric green, judge says not done → escalate
4. `signature == previous_signature` → escalate (not converging)
5. money budget spent → escalate
6. iteration budget spent → escalate
7. else → retry, `final_attempt` if the next one is last

`signature` is the tuple of **failed row names**, not the wording. Two equal signatures mean the last attempt changed nothing.

Commands. Live first.

`task loop:implementer` is gone from the root Taskfile. `loops/` was deleted on purpose.

Saturday self-check:

```
task test    # python3 -c "import harness; print('ok')"
```

To actually run the eight-step loop, use the Deep Agents port (after class, or as the instructor demo):

```
cd ../../solutions/sol2_implementer_deep_agents
python3 harness.py --repo ../../work/northwind-field-crm --ticket T001 --doer reference
python3 harness.py --repo ../../work/northwind-field-crm --ticket T001 --doer none
```

The lab README verify is `task test`. Demo from the Deep Agents port.

Expected results

`--doer none` writes nothing. The red gate must escalate:

```
gate: escalate
reason: red gate: no new test was observed failing. A test that passes before any code exists proves nothing.
```

If this run were green, the harness would be lying.

`--doer reference` copies `known-good` into `tests/**` then `app/**`, each phase bound by that role's WriteScope. Expect ten PASS rows and `gate: pass`.

Receipt. Three claims or nothing

`scripts/receipt.py` writes `.harness/receipt.json`.

1. Suite passed (exit 0 **and** readable junit **and** no failures)
2. Ran against **this tree** (`tree_hash` of tracked plus untracked)
3. Ran **after** the newest source edit

A zero exit code with no test report is the silent-skip bug wearing a green shirt.

The push-gate refusal you hit live lives on the CRM clone as a Claude Code hook, not under `labs/lab2_implementer/.claude/`. Read the refusal. It is the lesson.

Troubleshooting

SYMPTOM	LIKELY CAUSE	RESOLUTION
Computing ten rows by hand	stall on `score_attempt`	`return rubric.score(contract=contract, **evidence)`
Returning all failed ids	`red_gate` used `after.failed`	subtract `before` ids
Edited `loops/` or `rubric.py`	ignored the prompt	fill only `harness.py`
`task loop:implementer` missing	engine deleted	demo via `sol2_implementer_deep_agents/harness.py`
`--doer none` is green	red gate not stopping	empty new-ids must escalate
`import _root` fails	leftover TROUBLESHOOTING	lab stub does not import it

Fall behind

There is no drop-in `harness.py`. Watch Rick finish. Save first:

```
cp harness.py harness.py.my-attempt
```

See `FALL-BEHIND.md`. Restore with `git checkout -- harness.py`.

Take-home, not Saturday: issues 118. `labs/takehome/deep-agents/loop.py` and `labs/takehome/agent-sdk/loop.py` fill `build` and `run`. Those ports are a different runtime, not a drop-in for this stub.

Recap

What we built. Three functions that make a loop honest: new red ids, a ten-row rubric, Python-owned retries.

Takeaways

1. Two doers. Disjoint scope. The judge has no write method.
2. A test that passes before any code exists proves nothing.
3. "The tests passed" is one row of ten.
4. Same signature twice is stop.
5. A receipt proves three things, or it proves nothing.